# Parul Institute of Engineering & Technology

## Electronics & Communication Engineering Department

## ASSIGNMENT-4

### Subject Name: DIGITAL ELECTRONICS

### Subject Code: 303105220

**( Flip Flops, Counters and Registers )**

1. What is Flip flop? Explain with truth table the working of clocked RS flip-flop.
2. Draw and explain the working of following flip-flops
   [1] JK flip-flop [2] T-flip-flop [3] D-flip flop
3. State and Explain triggering methods used for flip-flop.
4. What is race-around condition in JK flip-flop?
5. Explain Master Slave Flip Flop through J.K Flip Flop with necessary logic diagram.
6. Design conversion logic to convert a JK Flip flop to T Flip flop.
7. Give classification of counters and explain asynchronous 4-bit binary ripple up counter. Explain the count sequence with waveform.
8. Design a synchronous counter
   a. 3-bit up/down synchronous counter using T-Flip flop.
   b. Mod-5 Synchronous up counter using JK-Flip flop.
   c. With the following binary sequence: 0,1,3,4,6 and repeat. Use T flip-flops.
9. Explain Ring Counter and Johnson's Counter with necessary logic diagram and waveform.
10. What is the function of shift register? With neat diagram explain the operation of 4-bit shift register.
    a. SISO ( Serial Input Serial Output )
    b. SIPO ( Serial Input Parallel Output)
    c. PISO ( Parallel Input Parallel Output)
    d. PIPO ( Parallel Input Parallel Output)